

Chapter 5: Interaction

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Chapter 4 introduced effect modification: the effect of a single treatment A varying across levels of another variable V . Many causal questions, however, concern two or more treatments applied together. Roughly, two treatments interact if the causal effect of one depends on the value we set for the other; Section 5.1 makes this precise.

This chapter defines interaction between two treatments in two frameworks:

- the counterfactual (potential outcomes) framework, which we have used so far;
- the sufficient-component-cause framework, which describes causal mechanisms.

This chapter is based on Hernán and Robins (2020, chap. 5, pp. 61-74). When joint interventions are feasible, knowing about interaction tells us which combination of interventions works best.

1 5.1 Interaction Requires a Joint Intervention (pp. 61-62)

1.1 Joint Interventions

Definition 1.1 (Joint Intervention and Joint Counterfactual). A **joint intervention** sets two or more treatments at once. For two treatments A and E , the **joint counterfactual outcome** $Y^{a,e}$ is the outcome that would have been observed had we intervened to set A to a and E to e .

Example 1.1 (Transplant and Vitamins: Four Joint Counterfactuals). Extend the heart transplant example with a second treatment, assigned before the first:

- E : multivitamin complex ($E = 1$) or no vitamins ($E = 0$);
- A : heart transplant ($A = 1$) or not ($A = 0$).

A joint intervention on A and E assigns everyone to one of four treatment combinations, so each individual has four joint counterfactual outcomes: $Y^{a=1,e=1}$, $Y^{a=1,e=0}$, $Y^{a=0,e=1}$, and $Y^{a=0,e=0}$.

1.2 Recursive Substitution

Remark 1.1 (Recursive Substitution). Suppose E is not affected by A (for example, because E is assigned before A), and that intervening to set E to the value it would have taken anyway does not change the outcome. Then an intervention on A alone leaves each individual's E at its actual value, so the counterfactual Y^a equals the joint counterfactual evaluated at that value:

$$Y^a = Y^{a,E}.$$

Combining this with consistency for A ($Y = Y^A$) gives $Y = Y^A = Y^{A,E}$: consistency is the special case of the same substitution in which a is also the actual value A .

Example 1.2 (Recursive Substitution for a Vitamin Taker). In Example 1.1, vitamins are assigned before transplant, so transplant cannot affect them. For an individual who actually took vitamins ($E = 1$), Remark 1.1 gives $Y^{a=1} = Y^{a=1,e=1}$: their outcome had they been transplanted equals their outcome had they been transplanted and given vitamins. For an individual who took no vitamins, $Y^{a=1} = Y^{a=1,e=0}$ instead.

Source: Hernán and Robins (2020, 61, margin note), which also points to Technical Point 6.2. If A could affect E , intervening on A alone would also change E , and the substitution would need the counterfactual value of E under a rather than its actual value.

1.3 Interaction in the Counterfactual Framework

Definition 1.2 (Interaction Between Two Treatments). Treatments A and E **interact** in bringing about an outcome Y if the causal effect of A on Y under a joint intervention that sets E to 1 differs from the causal effect of A on Y under a joint intervention that sets E to 0. Whether, and how, the two effects differ depends on the effect measure used to compare them.

Example 1.3 (Looking Up, Dressed or Naked (Hernán and Robins 2020, 61)). Chapter 2 asked whether your looking up at the sky makes other pedestrians look up too. Now randomize two treatments: whether you look up (A), and whether you stand in the street dressed or naked (E). If the effect of your looking up on other pedestrians when you are dressed differs from its effect when you are naked, then looking up and being dressed interact.

Definition 1.3 (Interaction on the Additive Scale). Let A and E be dichotomous treatments and Y a dichotomous outcome, and write $p_{ae} \stackrel{\text{def}}{=} \Pr[Y^{a,e} = 1]$ for the risk under the joint intervention that sets A to a and E to e . There is **interaction between A and E on the additive scale** in the population if the causal risk difference for A when everybody receives E differs from the causal risk difference for A when nobody receives E :

$$\Pr[Y^{a=1,e=1} = 1] - \Pr[Y^{a=0,e=1} = 1] \neq \Pr[Y^{a=1,e=0} = 1] - \Pr[Y^{a=0,e=0} = 1],$$

that is, $p_{11} - p_{01} \neq p_{10} - p_{00}$.

Example 1.4 (Transplant and Vitamins (Hernán and Robins 2020, 61–62)). Suppose the causal risk difference for transplant is

- 0.1 when everybody receives vitamins: $p_{11} - p_{01} = 0.1$;
- 0.2 when nobody receives vitamins: $p_{10} - p_{00} = 0.2$.

Because $0.1 \neq 0.2$, transplant and vitamins interact on the additive scale.

1.4 A and E Have Equal Status

Proposition 1.1 (Additive Interaction Is Symmetric in A and E). *With p_{ae} as in Definition 1.3, the causal risk difference for A differs between $e = 1$ and $e = 0$ if and only if the causal risk difference for E differs between $a = 1$ and $a = 0$:*

$$p_{11} - p_{01} \neq p_{10} - p_{00} \iff p_{11} - p_{10} \neq p_{01} - p_{00}.$$

Moreover, the two differences of risk differences are equal: $(p_{11} - p_{01}) - (p_{10} - p_{00}) = (p_{11} - p_{10}) - (p_{01} - p_{00})$.

i Proof

Proof. Both differences of risk differences equal the same **interaction contrast**:

$$\begin{aligned} (p_{11} - p_{01}) - (p_{10} - p_{00}) &= p_{11} - p_{01} - p_{10} + p_{00} \\ &= (p_{11} - p_{10}) - (p_{01} - p_{00}). \end{aligned}$$

So one is nonzero exactly when the other is. □

Example 1.5 (The Vitamin Effect Depends on Transplant). In Example 1.4, $(p_{11} - p_{01}) - (p_{10} - p_{00}) = 0.1 - 0.2 = -0.1$. By Proposition 1.1, $(p_{11} - p_{10}) - (p_{01} - p_{00}) = -0.1$ too: the causal risk difference for vitamins is 0.1 lower had everybody been transplanted than had nobody been transplanted (Hernán and Robins 2020, 62).

1.5 Additive and Multiplicative Scales

Proposition 1.2 (No Additive Interaction Means Additive Joint Effects (Technical Point 5.1)). *With p_{ae} as in Definition 1.3, there is no interaction between A and E on the additive scale if and only if the joint effect of both treatments is the sum of the effect of A alone and the effect of E alone:*

$$p_{11} - p_{00} = (p_{10} - p_{00}) + (p_{01} - p_{00}).$$

Proof

Proof. Subtracting the right-hand side from the left-hand side gives

$$\begin{aligned} (p_{11} - p_{00}) - [(p_{10} - p_{00}) + (p_{01} - p_{00})] &= p_{11} - p_{10} - p_{01} + p_{00} \\ &= (p_{11} - p_{01}) - (p_{10} - p_{00}). \end{aligned}$$

The displayed equality holds exactly when this difference is zero, which is exactly when $p_{11} - p_{01} = p_{10} - p_{00}$, that is, when there is no additive interaction. $\square$

Definition 1.4 (Superadditive and Subadditive Interaction (Technical Point 5.1)). With p_{ae} as in Definition 1.3, additive interaction between A and E is

- **superadditive** if $p_{11} - p_{00} > (p_{10} - p_{00}) + (p_{01} - p_{00})$;
- **subadditive** if $p_{11} - p_{00} < (p_{10} - p_{00}) + (p_{01} - p_{00})$.

Example 1.6 (Transplant and Vitamins Interact Subadditively). In Example 1.4, the proof of Proposition 1.2 shows that $(p_{11} - p_{00}) - [(p_{10} - p_{00}) + (p_{01} - p_{00})] = (p_{11} - p_{01}) - (p_{10} - p_{00}) = -0.1 < 0$, so the interaction is subadditive.

Definition 1.5 (Interaction on the Multiplicative Scale (Technical Point 5.1)). With p_{ae} as in Definition 1.3 and $p_{00} > 0$, there is **interaction between A and E on the multiplicative scale** if the causal risk ratio for both treatments is not the product of the causal risk ratios for each alone:

$$\frac{p_{11}}{p_{00}} \neq \frac{p_{10}}{p_{00}} \times \frac{p_{01}}{p_{00}}.$$

The interaction is **supermultiplicative** if the left side is greater and **submultiplicative** if it is smaller.

Example 1.7 (Interaction on One Scale but Not the Other). Suppose $p_{00} = 0.1$, $p_{10} = 0.2$, $p_{01} = 0.3$, and $p_{11} = 0.4$. On the additive scale, $p_{11} - p_{00} = 0.3$ and $(p_{10} - p_{00}) + (p_{01} - p_{00}) = 0.1 + 0.2 = 0.3$, so there is no additive interaction (Proposition 1.2). On the multiplicative scale, $p_{11}/p_{00} = 4$ while $(p_{10}/p_{00}) \times (p_{01}/p_{00}) = 2 \times 3 = 6$, so the interaction is submultiplicative.

Name the Scale

“ A and E interact” is incomplete without a scale: as Example 1.7 shows, the same four risks can show no interaction on one scale and interaction on another.

1.6 Interaction versus Effect Modification

i Recall: Effect Modification (Chapter 4)

A variable V that is not affected by A modifies the effect of A on Y on the additive scale if $E[Y^{a=1} - Y^{a=0} \mid V = 1] \neq E[Y^{a=1} - Y^{a=0} \mid V = 0]$ (Chapter 4^a). The definition uses only the counterfactuals Y^a for interventions on A .

^a04-effect-modification.qmd#def-effect-modification

Table 1: Effect modification versus interaction

	Effect modification by V	Interaction between A and E
Counterfactuals involved	Y^a	$Y^{a,e}$
Interventions	on A only	joint, on A and E
Status of the two variables	unequal: V is not intervened on	equal

Remark 1.2 (Unequal versus Equal Status). Effect modification concerns the causal effect of A only. In Chapter 4, sex modified the effect of transplant, but the effect of sex on death was never considered, so V and A were not on an equal footing. Interaction concerns the joint causal effect of A and E , and Proposition 1.1 shows that its definition treats A and E symmetrically (Hernán and Robins 2020, 62).

2 5.2 Identifying Interaction (pp. 62-64)

2.1 Identification for the Joint Treatment

Proposition 2.1 (Identifying Interaction by Standardization). *Let A and E be dichotomous treatments, Y a dichotomous outcome, and L a discrete vector of measured covariates. Suppose that, for every $a, e \in \{0, 1\}$:*

1. **Conditional exchangeability for the joint treatment:** $Y^{a,e} \perp\!\!\!\perp (A, E) \mid L$.
2. **Positivity:** $\Pr[A = a, E = e \mid L = l] > 0$ for every l with $\Pr[L = l] > 0$.
3. **Consistency:** $Y = Y^{a,e}$ for every individual with $A = a$ and $E = e$.

Then each joint counterfactual risk is identified by standardization,

$$p_{ae} = \Pr[Y^{a,e} = 1] = \sum_l \Pr[Y = 1 \mid A = a, E = e, L = l] \Pr[L = l],$$

so the contrasts in Definition 1.3 and Definition 1.5 are identified too.

i Proof

Proof. For each a and e , summing over the l with $\Pr[L = l] > 0$:

$$\begin{aligned}
 \Pr[Y^{a,e} = 1] &= \sum_l \Pr[Y^{a,e} = 1 \mid L = l] \Pr[L = l] && \text{(law of total probability)} \\
 &= \sum_l \Pr[Y^{a,e} = 1 \mid A = a, E = e, L = l] \Pr[L = l] && \text{(exchangeability and positivity)} \\
 &= \sum_l \Pr[Y = 1 \mid A = a, E = e, L = l] \Pr[L = l] && \text{(consistency)}.
 \end{aligned}$$

Positivity makes each conditional probability in the second and third lines well defined. $\square$

💡 Treat the Pair as One Treatment

Proposition 2.1 is the single-treatment result of earlier chapters applied to the combined treatment AE with four levels (11, 01, 10, 00). Standardization or IP weighting for that four-level treatment estimates all four p_{ae} , and any interaction contrast is then a contrast of those four risks.

Remark 2.1 (Sufficient, Not Necessary). The conditions of Proposition 2.1 are **sufficient** for identifying interaction. Part III of the book shows that they are not necessary when the two treatments occur at different times (Hernán and Robins 2020, 64). After this chapter, most of Parts I and II return to a single treatment A .

2.2 When E Is Randomized, Interaction Is Effect Modification by E

Proposition 2.2 (Randomized E : Interaction Equals Effect Modification by E). *Let A and E be dichotomous treatments and Y a dichotomous outcome. Suppose that*

1. *E is randomly and unconditionally assigned, so that $Y^{a,e} \perp\!\!\!\perp E$ for every a and e , and $\Pr[E = e] > 0$ for $e = 0, 1$;*
2. *recursive substitution holds: $Y^a = Y^{a,E}$ (Remark 1.1).*

Then $\Pr[Y^{a,e} = 1] = \Pr[Y^a = 1 \mid E = e]$ for every a and e . Hence there is interaction between A and E on the additive scale (Definition 1.3) if and only if E modifies the effect of A on the additive scale:

$$\Pr[Y^{a=1} = 1 \mid E = 1] - \Pr[Y^{a=0} = 1 \mid E = 1] \neq \Pr[Y^{a=1} = 1 \mid E = 0] - \Pr[Y^{a=0} = 1 \mid E = 0].$$

i Proof

Proof. For each a and e :

$$\begin{aligned} \Pr[Y^{a,e} = 1] &= \Pr[Y^{a,e} = 1 \mid E = e] && \text{(exchangeability for } E: Y^{a,e} \perp\!\!\!\perp E) \\ &= \Pr[Y^{a,E} = 1 \mid E = e] && (Y^{a,E} \text{ is } Y^{a,e} \text{ for those with } E = e) \\ &= \Pr[Y^a = 1 \mid E = e] && \text{(recursive substitution, } Y^a = Y^{a,E}). \end{aligned}$$

Substituting these four equalities into the inequality of Definition 1.3 gives the displayed inequality. $\square$

Example 2.1 (Randomized Vitamins). Suppose vitamins are randomly assigned, as in Proposition 2.2, and the risks are those of Example 1.4. Then among those who received vitamins, the causal risk difference for transplant is $\Pr[Y^{a=1} = 1 \mid E = 1] - \Pr[Y^{a=0} = 1 \mid E = 1] = p_{11} - p_{01} = 0.1$, and among those who did not, it is $p_{10} - p_{00} = 0.2$. So vitamins modify the effect of transplant, and the Chapter 4 methods for detecting effect modification by V apply with V replaced by E (Hernán and Robins 2020, 62–63).

2.3 Effect Modification Without Interaction

Remark 2.2 (Effect Modification Needs Fewer Assumptions than Interaction). Suppose we are willing to assume exchangeability for A but not for E , for example when A is randomized and we estimate its effect within subgroups defined by E . Then we can generally assess **effect modification by E** but not **interaction between A and E** : the effect of A within each

stratum of E needs no identifying assumptions about E , whereas the joint counterfactual risks p_{ae} do. Chapter 4 wrote such variables as V , for which no identifying assumptions are made.

Example 2.2 (Nationality: Effect Modification Without Interaction). In Chapter 4, nationality V modified the effect of transplant A , but it was argued to be a **surrogate** effect modifier: a marker for a factor, such as quality of care, that does act on Y and does interact with A . Nationality itself does not act on Y , so it does not interact with A (“no action, no interaction”). So the effect of A can be modified by a variable that does not interact with A (Hernán and Robins 2020, 64).

The reverse, interaction between A and E without modification of the effect of A by E , is logically possible but probably rare, because it requires dual effects of A and exact cancellations (Hernán and Robins 2020, 64, margin note citing VanderWeele 2009b).

3 5.3 Counterfactual Response Types and Interaction (pp. 64-66)

! Assumptions for Sections 5.3 to 5.6

None of the results so far requires deterministic counterfactual outcomes. From here to the end of the chapter, counterfactual outcomes are **deterministic**, and treatments and outcome are **dichotomous** (Hernán and Robins 2020, 64).

3.1 One Treatment: Four Response Types

Definition 3.1 (Response Type). With deterministic counterfactual outcomes, an individual’s **response type** (or **response pattern**) is the list of their counterfactual outcomes under every value of the treatment: $(Y^{a=0}, Y^{a=1})$ for one dichotomous treatment A , and $(Y^{a=1,e=1}, Y^{a=0,e=1}, Y^{a=1,e=0}, Y^{a=0,e=0})$ for two dichotomous treatments A and E .

Table 2: Response types for one dichotomous treatment (Hernán and Robins 2020, Table 5.1, p. 64)

Type	$Y^{a=0}$	$Y^{a=1}$	Zeus’s family (Table 1.1)
Doomed	1	1	Artemis, Athena, Persephone, Ares
Helped	1	0	Hebe, Kronos, Poseidon, Apollo, Hermes, Dionysus
Hurt	0	1	Rheia, Leto, Aphrodite, Zeus, Hephaestus, Polyphemos
Immune	0	0	Demeter, Hestia, Hera, Hades

Example 3.1 (Response Types in Zeus’s Family). With one dichotomous treatment and a dichotomous outcome there are four response types, and all four occur in Zeus’s family from Chapter 1 (Table 2): the **doomed** die whatever the treatment, the **helped** die only if untreated, the **hurt** die only if treated, and the **immune** survive whatever the treatment.

3.2 Two Treatments: Sixteen Response Types

With two dichotomous treatments each individual has four counterfactual outcomes, so there are $2^4 = 16$ response types (Miettinen 1982).

Definition 3.2 (Individual Interaction Contrast). For an individual with deterministic counterfactual outcomes $Y^{a,e}$, the **individual interaction contrast** is

$$c \stackrel{\text{def}}{=} Y^{a=1,e=1} - Y^{a=0,e=1} - Y^{a=1,e=0} + Y^{a=0,e=0},$$

the individual's effect of A when E is set to 1 minus their effect of A when E is set to 0. It depends only on the response type. Because expectations are linear, its population mean is the interaction contrast of Proposition 1.1: $E[c] = p_{11} - p_{01} - p_{10} + p_{00}$.

Table 3: Response types for two dichotomous treatments; superscripts are (a, e) . The first five columns follow Hernán and Robins (2020, Table 5.2, p. 65); the last adds the individual interaction contrast c of Definition 3.2

Type	$Y^{1,1}$	$Y^{0,1}$	$Y^{1,0}$	$Y^{0,0}$	c
1	1	1	1	1	0
2	1	1	1	0	-1
3	1	1	0	1	1
4	1	1	0	0	0
5	1	0	1	1	1
6	1	0	1	0	0
7	1	0	0	1	2
8	1	0	0	0	1
9	0	1	1	1	-1
10	0	1	1	0	-2
11	0	1	0	1	0
12	0	1	0	0	-1
13	0	0	1	1	0
14	0	0	1	0	-1
15	0	0	0	1	1
16	0	0	0	0	0

Example 3.2 (Individual Contrasts of Types 7 and 8). An individual with $(Y^{1,1}, Y^{0,1}, Y^{1,0}, Y^{0,0}) = (1, 0, 0, 1)$ (type 7 in Table 3) has $c = 1 - 0 - 0 + 1 = 2$: transplant kills them if they take vitamins and saves them if they do not. An individual with $(1, 0, 0, 0)$ (type 8) has $c = 1$: they die only if given both treatments.

3.3 Types Without Interaction

Example 3.3 (Types 1 and 16: No Effect of Either Treatment). Type 1 dies under every joint treatment and type 16 survives under every joint treatment. If everyone were of type 1 or 16, the sharp causal null hypothesis would hold for the joint treatment (A, E) (Hernán and Robins 2020, 65).

Proposition 3.1 (Six Types Without Interaction). *Assume deterministic counterfactuals and dichotomous A , E , and Y . For individuals of types 1, 4, 6, 11, 13, and 16 in Table 3, the effect of each treatment does not depend on the other:*

- *types 1 and 16: neither treatment has an effect;*
- *type 4: dies only if given vitamins, and type 13: dies only if not given vitamins (E alone matters);*
- *type 6: dies only if transplanted, and type 11: dies only if not transplanted (A alone matters).*

If everyone in the population has one of these six types, there is no interaction between A and E on the additive scale.

i Proof

Proof. The last column of Table 3 shows $c = 0$ for each of the six types, so $c = 0$ for every individual. By Definition 3.2, $p_{11} - p_{01} - p_{10} + p_{00} = E[c] = 0$, which is no additive interaction (Proposition 1.1). $\square$

3.4 Additive Interaction Requires Interaction Types

Definition 3.3 (Interaction Classes). Greenland and Poole (1988) grouped the other ten response types of Table 3 into three classes:

1. those who develop the outcome under only **one** of the four joint treatments: types 8, 12, 14, 15;
2. those who develop the outcome under **two** joint treatments, with the effect of each treatment reversed across levels of the other: types 7, 10;
3. those who develop the outcome under **three** of the four joint treatments: types 2, 3, 5, 9.

Unlike the 16 types, the three classes do not change when the labels 0 and 1 of A or E are swapped.

Example 3.4 (Classifying Types 7 and 8). The type-8 individual of Example 3.2 dies only under $(a, e) = (1, 1)$, so belongs to class 1. The type-7 individual dies under $(1, 1)$ and $(0, 0)$: transplant kills them with vitamins and saves them without, and vitamins kill them with transplant and save them without, so they belong to class 2.

Proposition 3.2 (Additive Interaction and the Interaction Classes). *Assume deterministic counterfactuals and dichotomous A , E , and Y , and let π_k be the proportion of the population with response type k in Table 3. Then*

$$p_{11} - p_{01} - p_{10} + p_{00} = (\pi_3 + \pi_5 + \pi_8 + \pi_{15}) - (\pi_2 + \pi_9 + \pi_{12} + \pi_{14}) + 2(\pi_7 - \pi_{10}).$$

Consequently:

- *additive interaction implies that some individuals belong to at least one of the three classes of Definition 3.3;*
- *no additive interaction implies that either nobody belongs to these classes or the contributions of the types present cancel exactly (for example, equal proportions of types 7 and 10, or of types 8 and 12).*

i Proof

Proof. By Definition 3.2, $p_{11} - p_{01} - p_{10} + p_{00} = E[c] = \sum_{k=1}^{16} \pi_k c_k$, where c_k is the value of c for type k in the last column of Table 3. Collecting the nonzero c_k gives the display. The types with $c_k \neq 0$ are exactly the ten types of the three classes. If no one belongs to a class, every term is zero; so additive interaction requires some $\pi_k > 0$ in a class. $\square$

Example 3.5 (Cancellation of Interaction Types). Let a fraction q of the population be type 8, a fraction q be type 12, and the rest type 16. Reading the risks off Table 3: $p_{11} = q$ (type 8), $p_{01} = q$ (type 12), $p_{10} = 0$, $p_{00} = 0$. Then

$$\begin{aligned} p_{11} - p_{01} - p_{10} + p_{00} &= q - q - 0 + 0 \\ &= 0, \end{aligned}$$

so there is no interaction on the additive scale in the population, even though every type-8 and type-12 individual has an effect of A that depends on E .

⚠ No Population Interaction Does Not Mean No Individual Interaction

As Example 3.5 shows, the absence of additive interaction in the population does not rule out individuals whose effect of A depends on E .

Miettinen’s 16 types are not invariant to switching the labels “0” and “1”; Greenland and Poole’s three classes are (Hernán and Robins 2020, 65, margin note). Greenland, Lash, and Rothman (2008) discuss such cancellations further.

Exercise 3.1 (Computing a Population Interaction Contrast). In a population, 30% are of type 2, 10% are of type 3, and the rest are of type 16. Is there additive interaction between A and E , and if so, is it superadditive or subadditive?

Solution

Solution. By Proposition 3.2, $p_{11} - p_{01} - p_{10} + p_{00} = \pi_3 - \pi_2 = 0.1 - 0.3 = -0.2$. This is nonzero, so there is additive interaction. It is negative, so by the proof of Proposition 1.2 the interaction is subadditive (Definition 1.4).

3.5 Monotonicity

Definition 3.4 (Monotonic Effects (Technical Point 5.2)). For one dichotomous treatment A , the causal effect of A on Y is **monotonic** if $Y^{a=1} \geq Y^{a=0}$ for every individual, that is, if nobody is helped by treatment.

For two dichotomous treatments A and E , the causal effects of A and E on Y are **monotonic** if every individual’s $Y^{a,e}$ is nondecreasing in both a and e . Equivalently, nobody has

- $Y^{a=1,e=1} = 0$ and $Y^{a=0,e=1} = 1$;
- $Y^{a=1,e=1} = 0$ and $Y^{a=1,e=0} = 1$;
- $Y^{a=1,e=0} = 0$ and $Y^{a=0,e=0} = 1$;
- $Y^{a=0,e=1} = 0$ and $Y^{a=0,e=0} = 1$.

Example 3.6 (Which Types Are Monotonic?). In Zeus’s family (Table 2), six individuals are helped by transplant, so the effect of transplant is not monotonic there. For two treatments, checking each row of Table 3 against the four forbidden pairs leaves only types 1, 2, 4, 6, 8, and 16: monotonic effects mean that everyone has one of these six types.

Source: Hernán and Robins (2020, Technical Point 5.2, p. 66).

3.6 Detecting Synergistic Types

Proposition 3.3 (Conditions for Types 7 and 8 to Exist (Fine Point 5.1)). Assume deterministic counterfactuals and dichotomous A , E , and Y , with p_{ae} as in Definition 1.3.

1. If $p_{11} - p_{01} - p_{10} > 0$, then some individuals have $Y^{a=1,e=1} = 1$ and $Y^{a=0,e=1} = Y^{a=1,e=0} = 0$ (types 7 or 8 in Table 3).

2. If, in addition, the effects of A and E are monotonic (Definition 3.4), then superadditive interaction, $p_{11} - p_{01} > p_{10} - p_{00}$, implies that some individuals are of type 8.

i Proof

Proof. For part 1, an individual with $Y^{a=1,e=1} = 1$ either has $Y^{a=0,e=1} = 1$, or has $Y^{a=1,e=0} = 1$, or is of type 7 or 8. So

$$\begin{aligned} p_{11} &\leq \Pr[Y^{a=0,e=1} = 1] + \Pr[Y^{a=1,e=0} = 1] + \Pr[\text{type 7 or 8}] \\ &= p_{01} + p_{10} + \Pr[\text{type 7 or 8}], \end{aligned}$$

and $p_{11} - p_{01} - p_{10} > 0$ forces $\Pr[\text{type 7 or 8}] > 0$.

For part 2, under monotonicity only types 1, 2, 4, 6, 8, and 16 occur (Example 3.6), and their contrasts in Table 3 give $p_{11} - p_{01} - p_{10} + p_{00} = \pi_8 - \pi_2$ (Proposition 3.2). Superadditive interaction makes this positive (proof of Proposition 1.2), so $\pi_8 > \pi_2 \geq 0$. $\square$

Example 3.7 (Checking for Types 7 and 8 in a Trial). Suppose a randomized experiment on both A and E , in which the conditions of Proposition 2.1 hold, gives $p_{11} = 0.5$, $p_{01} = 0.2$, and $p_{10} = 0.1$. Then $p_{11} - p_{01} - p_{10} = 0.2 > 0$, so by Proposition 3.3 some individuals would die if given both treatments but not if given either one alone.

⚠ Sufficient, Not Necessary

Both conditions in Proposition 3.3 are sufficient but not necessary. The first is so strong that it can fail in most populations in which types 7 or 8 exist (Hernán and Robins 2020, Fine Point 5.1, p. 67).

VanderWeele and Robins (2007a, 2008) developed this theory of sufficient cause interaction for two and three treatments. The monotonic-case result was reported by Greenland and Rothman and appears in Greenland, Lash, and Rothman (2008).

Exercise 3.2 (The Weaker Condition Under Monotonicity). Suppose the effects of A and E are monotonic and $p_{00} = 0.1$, $p_{10} = 0.2$, $p_{01} = 0.3$, and $p_{11} = 0.45$. Does part 1 of Proposition 3.3 apply? Does part 2?

Solution

Solution. Part 1 does not apply: $p_{11} - p_{01} - p_{10} = 0.45 - 0.3 - 0.2 = -0.05$, which is not positive. Part 2 does: $p_{11} - p_{01} = 0.15 > 0.1 = p_{10} - p_{00}$, so the interaction is superadditive, and under monotonicity some individuals are of type 8. In fact $\pi_8 - \pi_2 = 0.45 - 0.3 - 0.2 + 0.1 = 0.05$, so at least 5% are of type 8.

4 5.4 Sufficient Causes (pp. 66-69)

The variety of response types shows that A is not the only determinant of Y . The sufficient-component-cause framework represents the other determinants explicitly.

Definition 4.1 (Background Factor). In the sufficient-component-cause framework, a **background factor** is a dichotomous variable U , other than the treatments, that helps determine the outcome. By definition, background factors cannot be intervened on and are not affected by treatment (Hernán and Robins 2020, 67, margin note).

Example 4.1 (Allergy to Anesthesia). In an oversimplified version of the heart transplant example, suppose a transplant kills only individuals who are allergic to anesthesia. The indicator U_1 of allergy to anesthesia is a background factor: we cannot intervene on it, and transplant does not change it.

Definition 4.2 (Minimal Sufficient Cause). A **sufficient cause** of an outcome is a set of conditions, on the treatments and the background factors, whose joint presence inevitably produces the outcome. It is **minimal** if no proper subset of its conditions is itself sufficient. Its conditions are its **component causes**; “sufficient-component causes” refers to the sufficient causes and their components together.

Example 4.2 (Three Sufficient Causes for One Treatment (Hernán and Robins 2020, 66–67)). Continue Example 4.1. Suppose also that not receiving a transplant kills only those with an ejection fraction below 20% ($U_2 = 1$), and that pancreatic cancer at study start ($U_0 = 1$) kills whatever the treatment. Then death has three minimal sufficient causes:

- $A = 1$ together with $U_1 = 1$;
- $A = 0$ together with $U_2 = 1$;
- $U_0 = 1$, with no treatment component.

Table 4: The three sufficient causes of death in Example 4.2

Sufficient cause	Treatment component	Background factor
$A = 1$ with $U_1 = 1$	transplant	allergy to anesthesia
$A = 0$ with $U_2 = 1$	no transplant	ejection fraction below 20%
$U_0 = 1$	none	pancreatic cancer at study start

Figure 5.1 of the book draws each sufficient cause as a circle (“causal pie”) divided into its components.

i Fine Point 5.2: From Response Types to Component Causes

In Example 4.2, each combination of background factors determines one response type:

Table 5: Mapping between response types and sufficient-component causes for one treatment (Hernán and Robins 2020, Fine Point 5.2, p. 70)

Type	$Y^{a=0}$	$Y^{a=1}$	Component causes
Doomed	1	1	$U_0 = 1$ or $\{U_1 = 1 \text{ and } U_2 = 1\}$
Helped	1	0	$U_0 = 0, U_1 = 0, U_2 = 1$
Hurt	0	1	$U_0 = 0, U_1 = 1, U_2 = 0$
Immune	0	0	$U_0 = 0, U_1 = 0, U_2 = 0$

Each combination of component causes gives exactly one response type, but a response type can arise from several combinations (e.g., “doomed” arises from any combination with $U_0 = 1$, or with $U_1 = 1$ and $U_2 = 1$).

Remark 4.1 (Exchangeability in Terms of Component Causes). In the setting of Example 4.2, exchangeability $Y^a \perp\!\!\!\perp A$ for a dichotomous treatment and outcome means $\Pr[Y^{a=1} = 1 \mid A = 1] = \Pr[Y^{a=1} = 1 \mid A = 0]$ and $\Pr[Y^{a=0} = 1 \mid A = 1] = \Pr[Y^{a=0} = 1 \mid A = 0]$. By Table 5,

- those with $Y^{a=1} = 1$ are the doomed and the hurt, that is, those with $U_0 = 1$ or $U_1 = 1$;

- those with $Y^{a=0} = 1$ are the doomed and the helped, that is, those with $U_0 = 1$ or $U_2 = 1$.

So exchangeability holds exactly when $\Pr[U_0 = 1 \text{ or } U_1 = 1 \mid A = 1] = \Pr[U_0 = 1 \text{ or } U_1 = 1 \mid A = 0]$ and $\Pr[U_0 = 1 \text{ or } U_2 = 1 \mid A = 1] = \Pr[U_0 = 1 \text{ or } U_2 = 1 \mid A = 0]$ (Hernán and Robins 2020, Fine Point 5.2, p. 70).

See Greenland and Brumback (2002), Flanders (2006), and VanderWeele and Hernán (2006). VanderWeele and Robins (2008) generalized some results to two or more treatments.

4.1 Effect Size Depends on Background Factors

Example 4.3 (Prevalence of Allergy and the Size of the Effect). Take the sufficient causes of Example 4.2, and compare two populations that have the same joint distribution of U_0 and U_2 , with $\Pr[U_0 = 0] > 0$, but in which allergy to anesthesia ($U_1 = 1$) has prevalence

- 1% in the first;
- 10% in the second.

In each population, let U_1 be independent of (U_0, U_2) . Since $Y^{a=1} = 1$ exactly when $U_0 = 1$ or $U_1 = 1$, the risk had everyone been transplanted is $\Pr[Y^{a=1} = 1] = \Pr[U_0 = 1] + \Pr[U_0 = 0] \Pr[U_1 = 1]$, while $\Pr[Y^{a=0} = 1] = \Pr[U_0 = 1 \text{ or } U_2 = 1]$ is the same in both populations. So the causal risk difference is larger in the second population by $0.09 \Pr[U_0 = 0]$: the sufficient cause “ $A = 1$ plus $U_1 = 1$ ” is ten times more common there. A randomized experiment in each population, assigning half to $A = 1$, would estimate this larger effect.

This is the Chapter 4 point that the magnitude of a causal effect depends on the distribution of effect modifiers, made visible with sufficient-component causes (Hernán and Robins 2020, 68).

4.2 Two Treatments: Nine Sufficient Causes

Remark 4.2 (Nine Possible Sufficient Causes for Two Treatments). With dichotomous treatments A and E , there are 9 possible sufficient causes (Greenland and Poole 1988). Their treatment components are

1. $A = 1$ only;
2. $A = 0$ only;
3. $E = 1$ only;
4. $E = 0$ only;
5. $A = 1$ and $E = 1$;
6. $A = 1$ and $E = 0$;
7. $A = 0$ and $E = 1$;
8. $A = 0$ and $E = 0$;
9. neither A nor E .

Each also contains background factors, from $U_1, \dots, U_8$ and U_0 (Figure 5.2 of the book).

Example 4.4 (Sufficient Causes That Are Absent). Not all 9 sufficient causes need be present. If vitamins ($E = 1$) never kill anyone, whatever A is, then the 3 sufficient causes with component $E = 1$ are absent. If they were present, some individuals (e.g., those with $U_3 = 1$) would be killed by vitamins, that is, saved by withholding them (Hernán and Robins 2020, 69).

5 5.5 Sufficient Cause Interaction (pp. 69-71)

The counterfactual definition of interaction (Definition 1.3) is a contrast of counterfactual risks. It can be identified in an ideal randomized experiment on A and E without any knowledge of the mechanisms by which the treatments act. A second concept of interaction refers to mechanisms directly.

Definition 5.1 (Sufficient Cause Interaction). There is a **sufficient cause interaction** between treatments A and E in a population if some sufficient cause (Definition 4.2) has a component involving A and a component involving E , and at least one individual has all of its background-factor components, so that it would be completed under some joint intervention on A and E .

Example 5.1 (Background Factor U_5). Suppose individuals with $U_5 = 1$ develop the outcome when receiving both vitamins and transplant but not when receiving only one of them. Then a sufficient cause interaction exists if anyone has $U_5 = 1$. Hence, if some individual has $Y^{a=1,e=1} = 1$ and $Y^{a=0,e=1} = Y^{a=1,e=0} = 0$, a sufficient cause interaction between A and E is present (Hernán and Robins 2020, 69–70).

5.1 Synergism and Antagonism

Definition 5.2 (Synergism and Antagonism). A sufficient cause interaction between A and E is

- **synergism** if $A = 1$ and $E = 1$ are components of the same sufficient cause;
- **antagonism** if $A = 1$ and $E = 0$, or $A = 0$ and $E = 1$, are components of the same sufficient cause.

Antagonism between A and E can be viewed as synergism between A and “no E ” (or between “no A ” and E).

Example 5.2 (Synergism and Antagonism Between Transplant and Vitamins). The sufficient cause of Example 5.1, with components $A = 1$, $E = 1$, and $U_5 = 1$, is a synergism between transplant and vitamins. If, instead, some individuals died only when transplanted without vitamins, a sufficient cause with components $A = 1$ and $E = 0$ would be present: an antagonism between transplant and vitamins.

Rothman (1976) described synergism and antagonism within the sufficient-component-cause framework (Hernán and Robins 2020, 71, margin note).

5.2 Detecting Synergism Without Knowing the Mechanisms

Proposition 5.1 (Detecting Synergism from Counterfactual Risks). *Assume deterministic counterfactuals and dichotomous A , E , and Y , and suppose the outcome is generated by sufficient causes: for every individual and every joint treatment (a, e) , $Y^{a,e} = 1$ if and only if at least one sufficient cause is completed under (a, e) . If either condition of Proposition 3.3 holds (the second one together with monotonic effects), then there is synergism between A and E (Definition 5.2).*

i Proof

Proof. By Proposition 3.3, some individual has $Y^{a=1,e=1} = 1$ and $Y^{a=0,e=1} = Y^{a=1,e=0} = 0$. Under $(a, e) = (1, 1)$, some sufficient cause is completed for this individual. That sufficient cause cannot contain $A = 0$ or $E = 0$, because those components are absent under $(1, 1)$. It

cannot lack a component involving A , because it would then also be completed under $(0, 1)$, giving $Y^{a=0, e=1} = 1$. It cannot lack a component involving E , because it would then also be completed under $(1, 0)$, giving $Y^{a=1, e=0} = 1$. So it contains both $A = 1$ and $E = 1$, which is synergism. $\square$

Example 5.3 (Synergism in a Trial). In Example 3.7, $p_{11} - p_{01} - p_{10} = 0.2 > 0$. By Proposition 5.1, there is synergism between A and E , even though nothing is known about the background factors involved.

This is not surprising, for two reasons (Hernán and Robins 2020, 71): response types correspond to sufficient causes (Fine Point 5.2), and the inequalities are sufficient but not necessary, so they can fail even when synergism exists.

i Fine Point 5.3: “Biologic” Interaction

Sufficient cause interaction is often called **biologic interaction** (Rothman et al. 1980), but it need not involve treatments acting on each other.

Definition 5.3 (Compositional Epistasis). In genetics, **compositional epistasis** between two genetic factors A and E means that some individuals have response type 8 in Table 3: they develop the outcome when both factors are present, and not otherwise.

Example 5.4 (Two Alleles (Hernán and Robins 2020, Fine Point 5.3, p. 71)). Let A and E indicate a deleterious mutation in each of the two alleles of a gene that produces an essential protein (VanderWeele and Robins 2007a). Infants with both mutations ($A = 1, E = 1$) lack the protein and die within a week; those with one or no mutation survive. These infants are of type 8, so there is compositional epistasis. There is also synergism, since a sufficient cause of death contains $A = 1$ and $E = 1$, yet the alleles arguably do not physically act on each other.

VanderWeele (2010a) reviews tests for compositional epistasis.

6 5.6 Counterfactuals or Sufficient-Component Causes? (pp. 71-74)

The two frameworks answer different questions:

Table 6: Two frameworks for causation

	Sufficient-component causes	Counterfactuals
Starts from	a particular effect	a particular cause or intervention
Central question	by what mechanisms the outcome came about	what the outcome would be under an intervention
Describes mechanisms	yes	no

For estimating average causal effects of hypothetical interventions, the subject of the book, the counterfactual framework is the natural one.

The sufficient-component-cause model starts from an outcome and looks for the events that could have produced it; the counterfactual model starts from one factor and looks at what changing

that factor would do to the outcome (Hernán and Robins 2020, 71). In philosophy the sufficient-component-cause framework goes back to Mackie (1965), whose **INUS condition** for Y is an **I**nsufficient but **N**ecessary part of a condition which is itself **U**nnecessary but exclusively **S**ufficient for Y . A counterfactual framework of causation was already hinted at by Hume (1748).

6.1 Strengths and Limitations of Sufficient-Component Causes

Remark 6.1 (Useful for Teaching, Limited for Data Analysis). Sufficient-component causes are useful for teaching, because they illustrate

- why the size of a causal effect depends on the distribution of background factors (effect modifiers), as in Example 4.3;
- how effect modification, interaction, and synergism relate.

They are of limited use for data analysis, because in its classical form the framework

- is deterministic;
- gives conclusions that depend on the coding of the outcome;
- is restricted to dichotomous treatments and outcomes;
- requires large amounts of data to study the fine distinctions it makes.

Extensions to stochastic settings and to categorical or ordinal treatments might widen its use: VanderWeele (2010b) extended it to 3-level treatments, and VanderWeele and Robins (2012) related stochastic counterfactuals to stochastic sufficient causes (Hernán and Robins 2020, 72). The counterfactual framework will likely remain the one most often used, and apparent alternatives such as causal diagrams and decision theory are essentially equivalent to it (Chapter 6).

i Fine Point 5.4: Attributable Fractions Do Not Add

Recall from Fine Point 3.5 that the excess fraction for a treatment is the share of observed cases that would not have occurred had everyone been untreated. Suppose the excess fraction is 75% for A and 50% for E . A joint intervention cannot prevent $75\% + 50\% = 125\%$ of cases: no intervention, single or joint, can prevent more than 100%.

Example 6.1 (Zeus Is Counted Twice). Suppose Zeus has background factor $U_5 = 1$ (and no other background factors), so by Example 5.1 he dies only if he receives both treatments, and suppose he received $A = 1$ and $E = 1$ and died. Had either treatment been withheld, he would not have been a case. So Zeus is among the cases prevented by setting $a = 0$ (the 75% for A) and among those prevented by setting $e = 0$ (the 50% for E): he is counted in both excess fractions, which is why they can sum to more than 100%.

Example 6.2 (Phenylketonuria: 100% Genetic and 100% Environmental). Intellectual disability due to phenylketonuria occurs in genetically susceptible individuals who eat certain foods. Removing those foods from the diet would prevent every case, and so would replacing the susceptibility genes. So the excess fraction is 100% for the foods and 100% for the genes. When A and E can be components of the same sufficient cause, asking what fraction of disease is attributable to each separately makes little sense (Hernán and Robins 2020, Fine Point 5.4, p. 72).

See Rothman, Greenland, and Lash (2008).

6.2 Monotonicity Rules Out Some Sufficient Causes

Remark 6.2 (Monotonicity Rules Out Some Sufficient Causes (Technical Point 5.3)). Suppose the effects of A and E are monotonic (Definition 3.4), and the outcome is generated by sufficient causes as in Proposition 5.1. Consider an individual for whom a sufficient cause containing $A = 0$ is completed under some $(a = 0, e)$, while no sufficient cause is completed under $(a = 1, e)$, for example someone whose only background factor is $U_2 = 1$. That individual would have $Y^{a=0,e} = 1$ but $Y^{a=1,e} = 0$: treating them would prevent their outcome, contradicting monotonicity. So under monotonicity, a sufficient cause containing $A = 0$ never makes the difference to anyone's outcome, and the same argument applies to $E = 0$. In this sense, monotonicity rules out the sufficient causes that contain $A = 0$ or $E = 0$.

Example 6.3 (Smoking and Physical Inactivity (Hernán and Robins 2020, Technical Point 5.3, pp. 73-74)). If smoking ($A = 1$) never prevents heart disease and physical inactivity ($E = 1$) never prevents heart disease, then, in the sense of Remark 6.2, no sufficient cause of heart disease contains “not smoking” ($A = 0$) or “physically active” ($E = 0$). Figure 5.3 of the book crosses out the sufficient causes excluded this way.

7 Summary

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- **Interaction** between two treatments A and E is defined by joint counterfactuals $Y^{a,e}$: on the additive scale, $\Pr[Y^{1,1} = 1] - \Pr[Y^{0,1} = 1] \neq \Pr[Y^{1,0} = 1] - \Pr[Y^{0,0} = 1]$, a definition symmetric in A and E ; a multiplicative version uses risk ratios.
 - **Identification** requires exchangeability, positivity, and consistency for the joint treatment (A, E) ; if E is randomized, interaction coincides with effect modification by E .
 - **Effect modification** can occur without interaction (surrogate modifiers).
 - With deterministic, dichotomous counterfactuals there are **16 response types**; additive interaction requires individuals of interaction types, but their presence can cancel out.
 - **Sufficient cause interaction** means A and E appear in the same sufficient cause; synergism can sometimes be detected from counterfactual risks alone (Proposition 5.1).
 - The counterfactual framework describes what would happen under interventions and is the one used in the rest of the book; sufficient-component causes describe mechanisms and are mainly a teaching tool.

Looking ahead: Chapter 6 introduces causal diagrams, which the book describes as essentially equivalent to the counterfactual framework.

8 References

Hernán, Miguel A, and James M Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC. <https://miguelhernan.org/whatifbook>.